Name: Udhayakumar
Reg. No.: 192424332
Course Code: DSA 0601
Course: Data Handling And Visualization

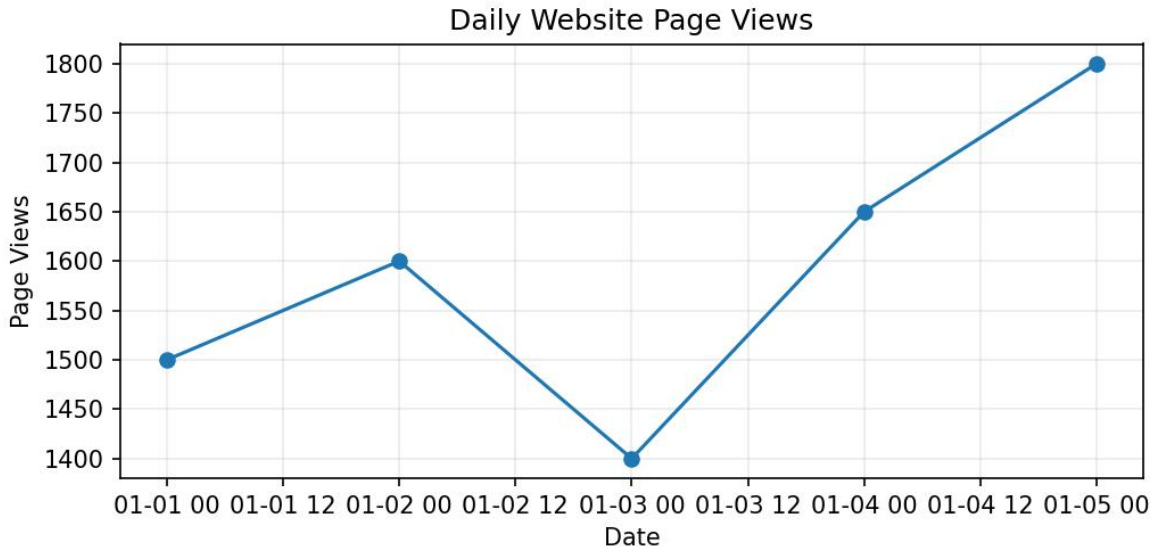
SET 25: WEBSITE TRAFFIC ANALYSIS

Q1. Tableau Line Chart

Aim: visualize daily page views over time.

Algorithm:
1. Import traffic data into Tableau.
2. Drag Date to Columns and Page Views to Rows.
3. Choose Line chart; add title and axis labels.

Output Visualization:



Result: Page views increase overall from 1,500 to 1,800, with a temporary drop on January 3.

Q2. R Bar Chart: Top N CTR

Aim: identify the three days with the highest click-through rate.

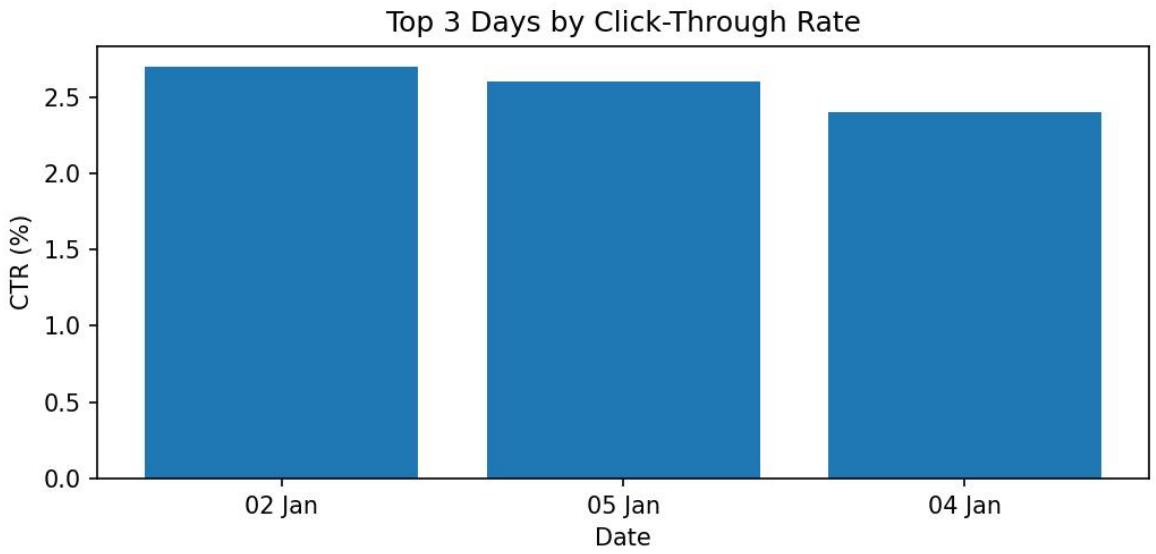
Algorithm:

1. Enter the dataset.
2. Sort CTR in descending order.
3. Select the top three days.
4. Draw a labeled bar chart.

R Code:

```
traffic <- data.frame(Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")), Page_Views=c(1500,1600,1400,1650,1800), CTR=c(2.3,2.7,2.0,2.4,2.6))
plot(traffic$Date, traffic$Page_Views, type="o", main="Daily Website Page Views", xlab="Date", ylab="Page Views")
top_n <- 3
top_days <- traffic[order(-traffic$CTR), ][1:top_n, ]
barplot(top_days$CTR, names.arg=format(top_days$Date,"%d %b"), col="steelblue", main="Top 3 Days by CTR", xlab="Date", ylab="CTR (%)")
print(top_days)
```

Output Visualization:



Result: January 2 has the highest CTR (2.7%), followed by January 5 (2.6%) and January 4 (2.4%).

Q3. R Stacked Area Chart

Aim: show likes, shares and comments over time.

Algorithm:

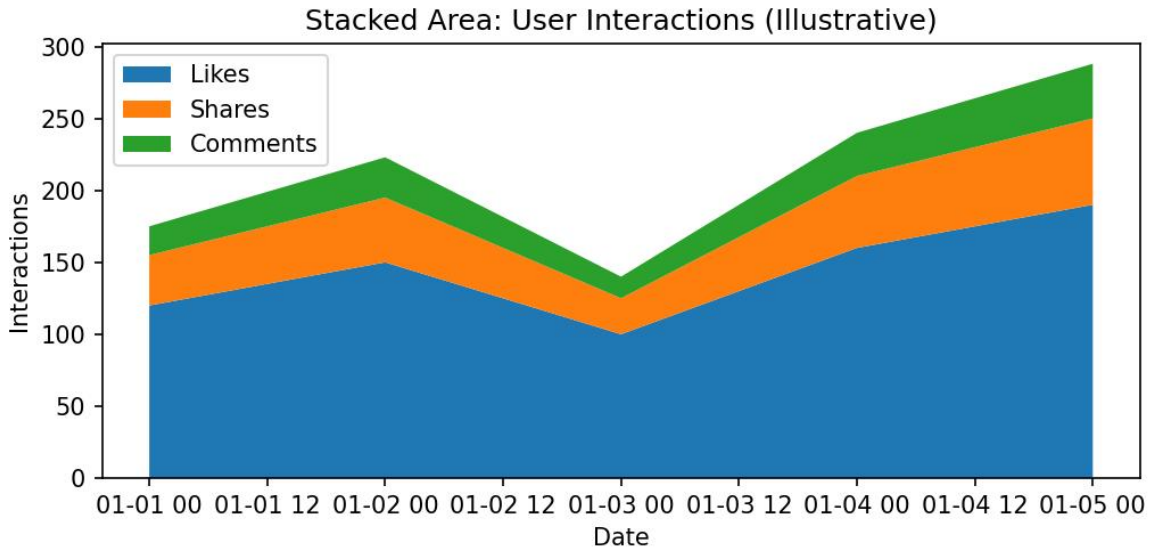
1. Enter interaction counts by date.
2. Stack the three interaction categories.
3. Add colors, legend and axis labels.

R Code:

```
interactions <- data.frame(Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03","2023-01-
04","2023-01-05")), Likes=c(120,150,100,160,190), Shares=c(35,45,25,50,60),
Comments=c(20,28,15,30,38))
# Illustrative counts; replace with real website data.
with(interactions, { y1<-Likes; y2<-Likes+Shares; y3<-Likes+Shares+Comments
 plot(Date,y3,type="n",main="Stacked Area of Interactions",xlab="Date",ylab="Interactions")
 polygon(c(Date,rev(Date)),c(rep(0,nrow(interactions)),rev(y1)),col="skyblue",border=NA)
 polygon(c(Date,rev(Date)),c(y1,rev(y2)),col="orange",border=NA)
 polygon(c(Date,rev(Date)),c(y2,rev(y3)),col="seagreen",border=NA)

legend("topleft",c("Likes","Shares","Comments"),fill=c("skyblue","orange","seagreen"),bty="
n") })
```

Output Visualization:



Result: Likes are the largest category in the illustrative sample. The supplied dataset lacks interaction counts, so replace the sample data with real values.

Q4. Tableau Interactive Dashboard

Aim: compare page views by traffic source and show their locations.

Algorithm:

1. Import data containing Traffic Source, Latitude, Longitude and Page Views.
2. Build a map using latitude/longitude and a bar chart by source.
3. Add both sheets to a dashboard.
4. Add a dashboard filter action and informative tooltips.

Result: The dashboard compares traffic sources geographically and by page views. The given dataset lacks source and location fields; these must be supplied separately.